

Compliance progress

1 Evidence-First Issuer Schema Reconciliation Pipeline

1.1 Motivation

Initial experiments showed that although rule extraction from SEBI ICDR regulations achieved high accuracy, the `maps_to` metadata used to infer issuer fields frequently misrepresented the true structure of issuer data in Red Herring Prospectuses (RHPs). In particular, multi-year financial quantities were often collapsed into scalar fields due to narrative-only extraction, leading to schema misalignment and verification failures.

1.2 Key Design Principle

To resolve this, we adopted an **evidence-first schema reconciliation** approach. Instead of treating the rule-derived schema as authoritative, the pipeline now infers issuer field types from *observed evidence* in the RHP, anchored explicitly to regulatory rules. This shifts schema design from a purely LLM-driven process to a data-supported, auditable workflow.

1.3 Pipeline Overview

The current pipeline consists of the following stages:

1. **Rule Extraction:** Atomic regulatory rules are extracted from the ICDR document, including structured `maps_to` hints for potential issuer fields.
2. **Provisional Schema Construction:** A conservative issuer schema is generated from `maps_to` fields using permissive types (e.g., `Option Nat`, `Option (List Nat)`, `Option Bool`) to avoid premature type commitment.
3. **Rule-Anchored Evidence Extraction:** For each rule and mapped field, the pipeline extracts supporting evidence from the RHP. Extraction is explicitly anchored to rule text and includes both narrative passages and structured financial tables.
4. **Table-Aware Parsing:** Financial tables in the RHP are deterministically extracted and normalized. Table-derived evidence (e.g., three-year financial series) is prioritized over narrative summaries when inferring field structure.
5. **Type Reconciliation:** Field types are inferred using deterministic heuristics based on observed evidence (parse success rates, list indicators, missingness). An optional LLM judge is invoked only for genuinely ambiguous cases.
6. **Schema Freezing:** The reconciled issuer schema is finalized and used as the basis for structured issuer extraction and Lean code generation.

1.4 Current Results

Applying this pipeline to a proof-of-concept RHP demonstrates substantial improvement:

- Multi-year financial quantities such as operating profits, net worth, and net tangible assets are correctly inferred as list-valued fields.
- Boolean compliance indicators are reliably distinguished from numeric quantities.
- Table-derived evidence eliminates ambiguity caused by narrative bullet summaries.
- The resulting issuer schema is stable, auditable, and better aligned with downstream formal verification requirements.

1.5 Status and Next Steps

The evidence-first reconciliation pipeline is now functionally complete and performs well on representative documents. Ongoing and future work includes:

- Extending table normalization to additional RHP formats.
- Improving evidence-to-rule linkage metadata for enhanced auditability.
- Integrating reconciled schemas into automated Lean rule and issuer instance generation.
- Evaluating schema stability across a broader corpus of RHPs.

Overall, the pipeline represents a shift from heuristic schema inference to a principled, data-driven approach that significantly improves robustness in regulatory compliance systems.